

2 Supplementary Information

3 SafeConf: registered-family error certificates for auditable single-cell perturbation prediction

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5 Supplementary Methods

6 S1. Scope and terminology

7 A **task** is one prediction vector paired with one experimental effect vector on the frozen 512-gene panel.
8 Depending on the source study, tasks correspond to donor–state combinations, technical groups, or
9 individual guides. A **target cluster** contains every task assigned to one held-out perturbation identity
10 within the relevant split. Calibration and coverage calculations keep the complete cluster together.

11 A **registered family** is a list of model checkpoints, seeds, output genes, normalization rules, and
12 aggregation rules fixed before evaluation truth is accessed. The main family contains five scGPT fits and
13 five GEARS fits with seeds 3407–3411. A family certificate describes this finite collection; it is not a
14 claim about every possible model trained with the same architectures.

15 For a task with predictions $p_1, \dots, p_m$ and experimental truth y , R_F is the root mean of the members’
16 squared RMSE values and W_F is the largest member RMSE. The **family lower certificate** D_F measures
17 RMS dispersion around the family centroid. The **worst-member lower certificate** $\Delta_F/2$ is half the
18 maximum pairwise prediction distance. The upper certificates combine a reference-centroid conformal
19 event with an observable centroid-shift penalty.

20 S2. Frozen study designs

Study	Train targets	Validation targets	Calibration targets	Evaluation targets	Evaluation tasks	Cluster content
Primary CD4	study-specific prospective panel	study-specific prospective panel	160 total, 40 in each donor rotation	640	1920	Rest, 8-h stimulation, and 48-h stimulation
Sunshine	54	10	30	50	400	eight gem_group technical groups
XuCao	65	32	29	27	73	all eligible guides for one target
GSE225807	28	9	19	20	40	two guides for one RBP target

21 The Primary CD4 count of 640 evaluation clusters comprises four donor rotations with 160 evaluation
 22 targets each. Donor-specific rotations are not presented as independent studies. In Sunshine, gem_group
 23 labels are technical rather than biological contexts. In XuCao, cell-cycle phase is observed after
 24 perturbation and therefore was not used as a pre-perturbation context variable. GSE225807 was the only
 25 study executed as a new, fully preregistered registered-family confirmation.

26 S3. Derivation of the family RMS identity

27 Let $\dot{p} = m^{-1} \sum_i p_i$. For one family member,

$$28 \quad \|p_i - y\|^2 = \|(p_i - \dot{p}) + (\dot{p} - y)\|^2.$$

29 Expanding the inner product gives

$$30 \quad \|p_i - \dot{p}\|^2 + \|\dot{p} - y\|^2 + 2\langle p_i - \dot{p}, \dot{p} - y \rangle.$$

31 Averaging over members removes the cross term because $m^{-1} \sum_i (p_i - \dot{p}) = 0$. Hence

$$32 \quad \frac{1}{m} \sum_i \|p_i - y\|^2 = \frac{1}{m} \sum_i \|p_i - \dot{p}\|^2 + \|\dot{p} - y\|^2.$$

33 With the definitions in the manuscript,

34
$$R_F^2 = D_F^2 + \| \hat{p} - y \|^2.$$

35 The gene-normalized Euclidean norm differs from the usual Euclidean norm by the constant $G^{-1/2}$, so the
36 same inner-product identity holds. The deterministic inequality $D_F \leq R_F$ follows because the remaining
37 term is non-negative.

38 This is the classical ambiguity decomposition for squared loss [1]. SafeConf uses the identity to define an
39 auditable lower certificate for a frozen heterogeneous prediction family; it does not claim the identity as a
40 new theorem.

41 **S4. Worst-member lower certificate**

42 Choose family members a and b attaining the diameter $\Delta_F = \| p_a - p_b \|$. The triangle inequality gives

43
$$\Delta_F \leq \| p_a - y \| + \| p_b - y \| \leq 2 \max_i \| p_i - y \|.$$

44 Therefore $\Delta_F/2 \leq W_F$. As with the family RMS lower bound, no calibration sample or target truth is
45 needed to calculate the left side.

46 **S5. Transfer of the reference upper event**

47 Let c be the frozen reference centroid and let the reference calibration event be $\| c - y \| \leq U$. The
48 registered-family centroid $\hat{p}$ may differ from c . Because

49
$$\| \hat{p} - y \| \leq \| c - y \| + \| \hat{p} - c \|,$$

50 the reference event implies $\| \hat{p} - y \| \leq U + s$, where $s = \| \hat{p} - c \|$. Combining this with the identity in S3
51 yields

52
$$R_F \leq \sqrt{(U + s)^2 + D_F^2}.$$

For each family member,

$$\|p_i - y\| \leq \|p_i - \hat{p}\| + \|\hat{p} - y\| \leq r_F + U + s,$$

which gives the worst-member upper certificate. The shift s is essential: omitting it would silently apply a guarantee calibrated for one centroid to a different prediction.

S6. Target-cluster split conformal rule

For target j , all associated tasks $t \in T_j$ contribute one score

$$S_j = \max_{t \in T_j} \{e_t - b(x_t)\},$$

where e_t is reference-centroid RMSE and $b(x_t)$ is a frozen non-negative base. With n calibration targets and nominal miscoverage α , the correction is the ordered score with rank

$$k = \text{ceil}((n+1)(1-\alpha)).$$

The target is covered only when every task in the target is below its task-level upper bound. This prevents two guides or repeated contexts of the same target from being counted as independent calibration examples. Under exchangeability of target clusters, the usual split-conformal marginal coverage argument applies [2]. The procedure does not establish conditional coverage for every molecular class or transport under arbitrary distribution shift.

S7. Reference bases and efficiency comparison

Study	Frozen reference base	Selection status
Primary CD4	donor-specific predicted-magnitude base	frozen before the registered-family reformulation
Sunshine	predicted magnitude plus model-disagreement terms	frozen before calibration
XuCao	constant, magnitude, magnitude-plus-lower, and ExtraTrees compared under the same cluster correction	adaptive method failed the registered efficiency gate
GSE225807	constant centroid bound only	fixed in the preregistration

In XuCao, the ExtraTrees model used 18 prediction-derived features and validation-target errors. It covered all 27 evaluation targets but produced a mean upper bound of 0.2315 RMSE, compared with 0.2168 for the constant comparator. The mean difference was +0.0147, with a target-cluster bootstrap 95% interval of +0.0107 to +0.0188. The adaptive method was therefore rejected before the GSE225807 design was finalized.

S8. Family-composition analyses

The registered ten-member family was compared with two architecture-level means: the mean of five scGPT predictions and the mean of five GEARS predictions. The seed-level family increased median lower-bound tightness by 0.0826 in Primary CD4, 0.0857 in Sunshine, and 0.1011 in XuCao. Every paired task favored the seed-level family.

A no-change prediction was added only as a post-evaluation negative-control stress test. It did not improve tightness in any XuCao task and changed the family centroid in every study. The upper-certificate transfer consequently paid a nonzero shift penalty. This analysis illustrates why member identities and seeds must be fixed independently of evaluation outcomes.

The family centroid calculated from all ten seed-level vectors agreed with the mean of the two five-seed architecture means to within 1.89×10^{-9} RMSE in the retrospective studies. This check detects inconsistent weighting or member loss.

S9. Fully preregistered GSE225807 confirmation

S9.1 Eligibility and split

Eligible RBP targets had exactly two assigned guides, at least five cells per guide, at least 20 cells across the two guides, and complete coverage on the frozen expression and vocabulary axes. Salted SHA-256

target identities were assigned within cell-count strata. The split contained 28 training targets, 9 validation targets, 19 calibration targets, and 20 evaluation targets.

Negative-guide controls contained 91 cells. The logical access record contained 1,348 supervised-training rows, 456 validation rows, 926 calibration rows, and 1,051 evaluation rows. Calibration and evaluation expression rows were absent from the pretruth scoring interface. Evaluation rows were opened only after the conformal threshold and all gates had been committed to both GitHub and Gitee.

S9.2 Registered gate

The 90% constant centroid threshold was the 18th ordered maximum centroid RMSE among 19 calibration targets:

$$U=0.18978567801751686.$$

The registered success rule required simultaneous family upper coverage for at least 17 of 20 evaluation targets, zero deterministic lower violations, ambiguity-identity residual no greater than 10^{-10} , and no access violation. The observed family result was 16/20; the registered overall status is therefore FAIL.

Four targets failed simultaneous family coverage: HNRNPC, DDX6, SLTM, and DDX42. DDX42 also failed the worst-member upper bound. All 40 guide tasks satisfied both deterministic lower bounds. The maximum identity residual was 2.26×10^{-17} .

S9.3 Finite-calibration reference calculation

For continuous exchangeable scores, selecting the 18th order statistic among 19 calibration scores gives a conditional future coverage probability distributed as $Beta(18, 2)$. Mixing a binomial count of 20 evaluation targets over this distribution gives

$$P(K \leq 16) = \sum_{k=0}^{16} \binom{20}{k} \frac{B(k+18, 20-k+2)}{B(18, 2)} = 0.186813.$$

This calculation explains the sampling variability induced by a small calibration set. It was computed after the failed registered gate and does not change the gate, the threshold, or the reported 16/20 outcome.

S10. Cross-study synthesis

Study	Tasks	Targets	Family lower violations	Family target upper coverage	Median family lower tightness
Primary CD4	1920	640	0	579/640 (90.47%)	0.240
Sunshine	400	50	0	44/50 (88.00%)	0.170
XuCao	73	27	0	27/27 (100%)	0.255
GSE225807	40	20	0	16/20 (80.00%)	0.513
Combined, descriptive	2433	737	0	666/737 (90.37%)	not pooled

The combined exact target interval is 88.00%–92.40%. It is reported descriptively because the synthesis combines different studies, task structures, bases, and analysis timing. It is not a new conformal calibration and cannot turn the GSE225807 gate into a success.

S11. Reproducibility and access chronology

The GSE225807 release was preserved in the following order:

Stage	Commit	Information fixed or opened
F1 metadata freeze	b4701ac	eligibility, salted identity split, task contract
F1 runtime hardening	fc0de74	implementation checks before truth release
F2 pretruth release	2f4f148	model family and prediction-derived geometry
F3 calibration lock	420e536	calibration truth and threshold
F4 final evaluation	593f663	one-time evaluation truth and registered outcome
post-F4 synthesis	744b5ef	retrospective four-study aggregation

The minimal-release validator recomputes task and target counts, deterministic inequalities, upper-coverage flags, identity residuals, the GSE225807 gate, and the beta-binomial probability from committed release tables. It ran 12,033 checks with zero failures under Python 3.9.25 and Python 3.12.4. A separate pre-submission integrity audit contained 18 checks, all passing.

S12. Model specificity and pretruth provenance

The following distinctions were fixed to prevent a family-level quantity from being interpreted as confidence in one model:

Question	Operational answer	Uses evaluation truth?	Reporting boundary
What does D_F certify?	RMS error across the complete registered family	no	does not certify any named member separately
What does $\Delta_F/2$ certify?	error of the worst member in the registered family	no	does not identify which member is worst
What is predicted magnitude?	RMSE of a predicted effect vector relative to the zero-effect vector	no	it is not the measured perturbation effect
Does a small scGPT-GEARS distance imply safety?	no	no	both architectures may share the same error
Can a cross-dataset upper bound be reported without target-domain calibration?	no	not applicable	only the deterministic lower certificate transfers algebraically

The five seed-level predictions were averaged within architecture before this diagnostic. In Primary CD4, scGPT and GEARS task errors had Spearman $\rho = 0.9747$, and their highest-error quintiles had Jaccard overlap 0.8916. In Sunshine, the corresponding values were 0.9924 and 0.9512. Their distance was associated with scGPT and GEARS RMSE at $\rho = -0.2112$ and -0.1739 in Primary CD4, and at 0.0586 and 0.0594 in Sunshine. Thus, error difficulty was largely shared, while the between-architecture distance was not a monotone single-model confidence score.

These correlations were calculated after truth release and serve only as interpretation checks. The deployment-time lower certificates use prediction vectors alone. The target expression used to calculate RMSE never enters the prediction-distance, predicted-magnitude, context-similarity, or training-support features.

S13. Difficulty ladder and direct-transfer stress test

The difficulty audit reused frozen prediction records; it did not select a predictor after viewing the new certificate result. Four multi-context matrices were included: Frangieh genetic perturbations, Lara ex vivo genetic perturbations, Santinha genetic perturbations, and Cui cytokine stimulation. For each matrix, two inductive reference predictors were evaluated under:

1. a random missing context–perturbation pair;
2. a complete unseen context row;
3. a complete unseen perturbation column;
4. a cell whose context and perturbation were both unseen.

Only 25%, 50%, 75%, or 100% of the permitted training submatrix was made available. Across the four fractions, each setting contributed 276, 879, 690, and 204 task instances, respectively, for a total of 8,196. Every task satisfied both deterministic inequalities. At full fraction, the perturbation-cluster bootstrap summary was:

Held-out structure	Macro median tightness	95% bootstrap interval
Random missing pair	0.328	0.315–0.342
Unseen context	0.260	0.252–0.269
Unseen perturbation	0.175	0.163–0.183
Context and perturbation both unseen	0.148	0.133–0.158

Bootstrap resampling kept every context associated with one perturbation identity together. Dataset medians were then averaged, preventing large matrices from dominating the summary. Repeated training fractions and folds were not counted as independent experiments.

The direct-transfer component added 553 sciPlex3-to-OpenProblems tasks and 28 sciPlex3-to-sciPlex4 tasks. Family-RMS and worst-member lower violations were both zero. Median family-RMS tightness was 0.703 and 0.641. The first ratio is high because the two reference predictors diverged while neither beat the zero-effect baseline, not because transfer was accurate. Target-domain calibration was absent, so no reference-centroid upper bound or cross-dataset coverage claim was constructed.

The full stress audit therefore contains 8,777 task instances. It extends the deterministic validity check beyond random missing pairs, while leaving the ten-member scGPT–GEARS four-study analysis as the main certificate evidence.

S14. Adaptive upper baselines and native PRESCRIBE comparator

S14.1 Nested target-level upper-baseline development

Primary CD4 and Sunshine were each resplit 50 times by complete perturbation identity into model-fitting, conformal-calibration, and evaluation partitions. Every candidate received the same target-level correction. The candidates were constant split conformal, predicted magnitude, magnitude plus the two-model lower bound, seed spread, Ridge, ExtraTrees, random forest, and 0.80-quantile gradient boosting.

Study	Method	Mean target coverage	Mean upper RMSE	Constant mean	Relative reduction	Paired repeat 2.5%–97.5%
Primary CD4	ExtraTrees	0.894	0.2062	0.2119	2.72%	–1.52%–7.39%
Sunshine	ExtraTrees	0.905	0.5389	0.5466	1.41%	–6.60%–7.72%

The resplits overlap and quantify development stability rather than independent replication. The selected ExtraTrees configuration was then frozen for XuCao, where it produced a wider upper bound than the constant method by 0.0147 RMSE (target-cluster bootstrap interval 0.0107–0.0188). This failed confirmation motivated the constant-only GSE225807 preregistration.

S14.2 PRESCRIBE at its own endpoint

PRESCRIBE was evaluated on 48 tasks from two Norman panels using its paper’s predictor-specific evidential confidence and prediction–truth correlation endpoint [3]. The two-panel macro association of combined confidence with Pearson effect accuracy was $\rho = 0.3170$ (bootstrap interval 0.0324–0.5669). Predicted magnitude reached $\rho = 0.3104$. Their paired difference was 0.0065 (–0.0317 to 0.0534), while the confidence score and predicted magnitude were almost rank-equivalent ($\rho = 0.9952$, 0.9746–0.9996).

This is a native-endpoint contextual comparator. It does not convert PRESCRIBE confidence into an upper or lower bound, and it does not compare the same estimand as SafeConf. The result is retained because it tests whether a named uncertainty method adds ordering information beyond a simple pretruth baseline in the setting for which it was designed.

Supplementary Data Files

The following files are included in the submission package:

- tables/Table_S1_family_comparisons.csv: paired registered-family composition comparisons;
- tables/Table_S2_gse225807_targets.csv: all 20 GSE225807 target-level outcomes;
- tables/Table_S3_gse225807_tasks.csv: all 40 GSE225807 guide-level certificates;
- tables/Table_S4_difficulty_setting_summary.csv: dataset-, fraction-, and setting-level difficulty summaries;
- tables/Table_S5_difficulty_macro_bootstrap.csv: perturbation-cluster bootstrap intervals for the difficulty ladder;
- tables/Table_S6_cross_dataset_summary.csv: direct-transfer certificate summaries and claim boundaries;
- tables/Table_S7_difficulty_task_certificates.csv: all 8,196 Cartesian stress-test tasks;
- tables/Table_S8_cross_dataset_task_certificates.csv: all 581 direct-transfer tasks;
- tables/Table_S9_model_error_concordance.csv: scGPT-GEARS shared-error diagnostics;
- tables/Table_S10_score_model_specificity.csv: score associations with model-specific and family error objects;
- tables/Table_S11_nested_upper_baselines.csv: 50-repeat target-level upper-baseline summary;
- tables/Table_S12_prescribe_native_endpoint.csv: PRESCRIBE native-endpoint associations;
- tables/Table_S13_prescribe_incremental.csv: paired increments over predicted magnitude;
- tables/Table_S14_prescribe_redundancy.csv: score-to-magnitude rank associations;
- tables/Table_2_certificate_results.csv: study-level certificate results used in the manuscript;
- figures/Figure_1_method_and_protocol.* through Figure_5_reproducibility_chain.*: 600-dpi PNG, vector PDF, and editable SVG exports.

203 The complete main-certificate release remains in docs/实验结果/E181_* through E186_*; secondary
204 audits are stored in E145_*, E178_*, E179_*, and E187_*. The source-data validator is invoked from the
205 repository root with:
206 python tools/scripts/validate_current_certificate_release.py

207 References

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